Lecture 5 deliverable
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1) The voltage drop without any capacitor was measured to be 3.4V.

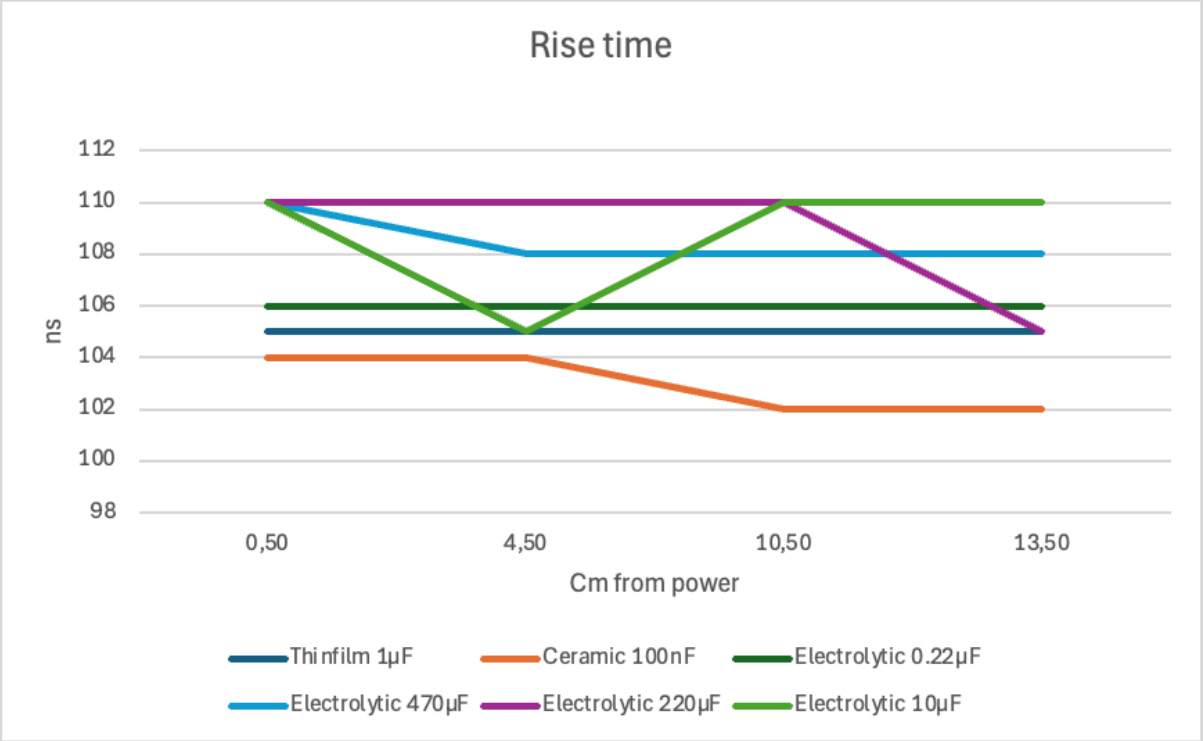
2) The most reduced voltage drop we saw was the thin film 1μF capacitor and the Electrolyte 470μF/220μF capacitors, they all dropped the pk-pk voltage to about 160 mV which is a decrease of 3.24V or a 95% drop compared to the no capacitor test.

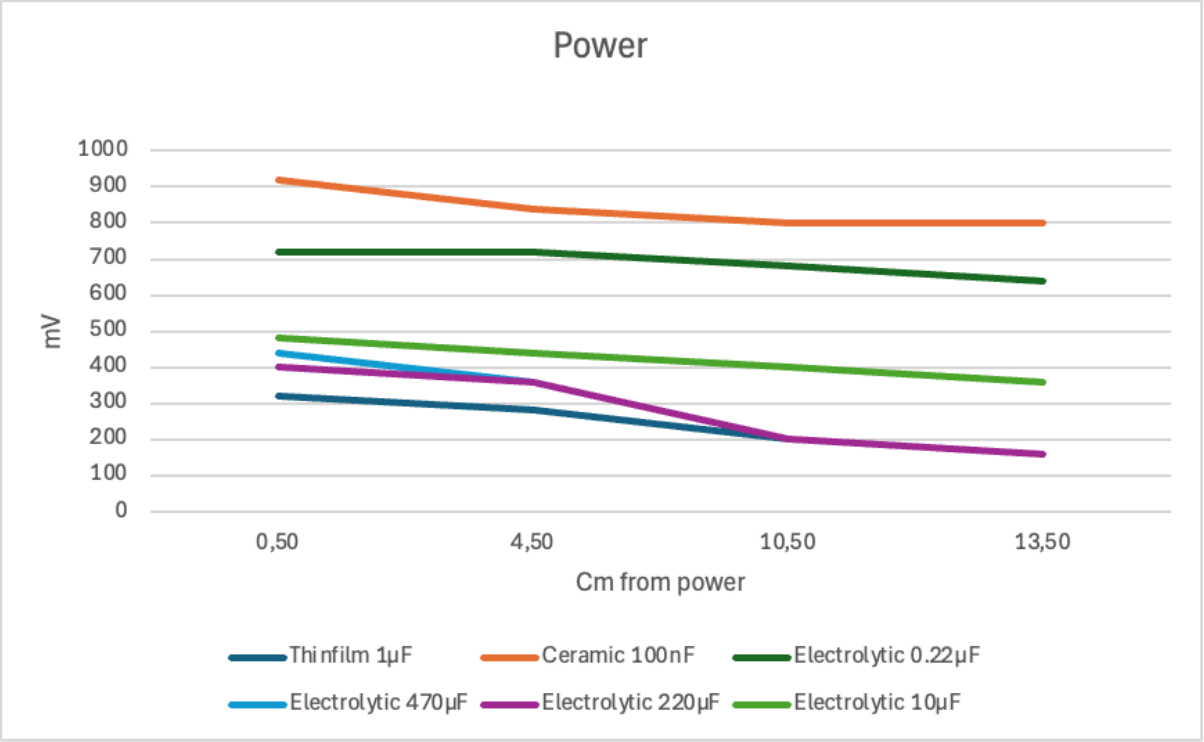
3) The distance between the capacitor and the load/IC is important as this makes the drop in voltage much lower and the rise time much lower, we could see that the closer you get to the load/IC the smaller the drop and rise time becomes.

4) The placement is the most important!!! However, if the capacitor is too small, then it will not be capable of supplying a sufficient amount of energy during a current spike therefore size still matters but to a lesser extent than proper placement. We could see this in action as we could compare the small and big capacitors. We here take an example: the small 10μ and the big 470μ capacitors at the side of the power have a drop of 480mV and 400mV showing that size matters, but much less than distance. When these two were put close to the IC/load we saw a drop of 360mV and 160mV showing that distance plays a very big role. While size still has a contribution. If we look at a better type of resistor, the thin film 1μ we see a smaller drop away from the IC but even smaller at the IC with 320mV at source and 160mV at the load. Showing that size matters way less than distance and type.

5) Pigtail instead of a probe ground, to minimize ground loop area and inductance, reducing noise and signal distortion in high-frequency measurements. If measurement is done using probe ground, you introduce a lot of unnecessary noise.

Plots of pk-pk voltage and rise time compared to distance and Capacitor type





Data table for distance and cap.

		nano sekunder	**Milivolt**
BJT	Cap: None	Risetime	Power
0.5cm	Close to power	240ns	3.4V

4.5cm	1/4 the way	X	X
10.5cm	3/4 the way	X	X
13.5cm	Close to GND	X	X
	Cap: Thinfilm 1µ	Risetime	Power
0.5cm	Close to power	105	320
4.5cm	1/4 the way	105	280
10.5cm	3/4 the way	105	200
13.5cm	Close to GND	105	160
	Cap: Cer 100n	Risetime	Power
0.5cm	Close to power	104	920
4.5cm	1/4 the way	104	840
10.5cm	3/4 the way	102	800
13.5cm	Close to GND	102	800
	Cap: Elec 0.22µ	Risetime	Power
0.5cm	Close to power	106	720
4.5cm	1/4 the way	106	720
10.5cm	3/4 the way	106	680
13.5cm	Close to GND	106	640
	Cap: Elec 470µ	Risetime	Power
0.5cm	Close to power	110	440
4.5cm	1/4 the way	108	360
10.5cm	3/4 the way	108	200
13.5cm	Close to GND	108	160
	Cap: Elec 220µ	Risetime	Power
0.5cm	Close to power	110	400
4.5cm	1/4 the way	110	360
10.5cm	3/4 the way	110	200
13.5cm	Close to GND	105	160
	Cap: Elec 10µ	Risetime	Power
0.5cm	Close to power	110	480
4.5cm	1/4 the way	105	440
10.5cm	3/4 the way	110	400
13.5cm	Close to GND	110	360

1) PCB 4+ was clearly the worst: 261 mV pk-pk with a square wave (vs. 40 mV on PCB 4 and 2 mV on the good board). This shows that return-path interruptions are the biggest sinner of crosstalk.

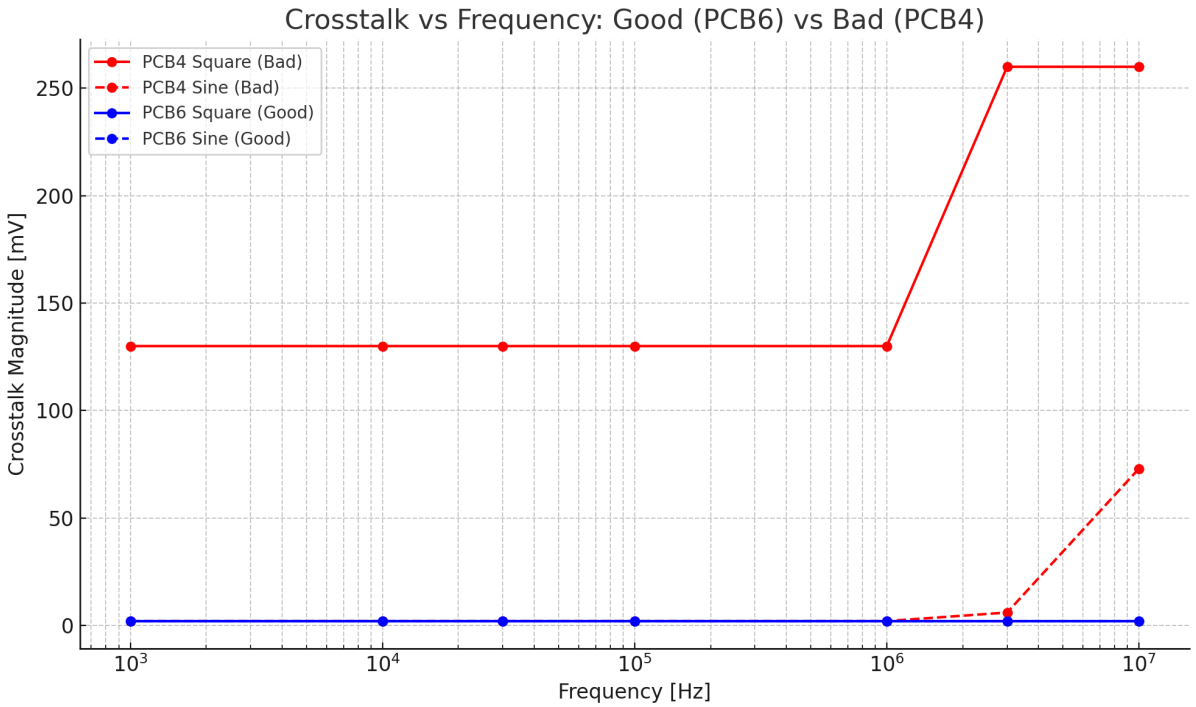
2) Did terminating the crossing trace in PCB 4+ help or worsen crosstalk? Why?
It made the crosstalk a lot worse. Adding 25 Ω terminations at both ends of the crossing trace effectively created a low-impedance, ground-referenced path right across the slot. That encourages larger return currents through the interrupted ground region and makes the crosstalk to the victim line worse, hence the jump from 40 mV (PCB 4) to 261 mV (PCB 4+). In other words, we made the mistake worse with more energy.

3) How do PCBs 5–7 (crossings on the same side or with jumper wires) perform compared to the solid ground plane? Why?
They matched the solid plane (2 mV, which comes from the test setup). Moving the crossing to the top layer with jumpers/0 Ω resistors keeps the bottom ground plane solid, so the return current stays directly under the aggressor trace, minimizing loop area and both capacitive and inductive coupling. That's why PCBs 5, 6, 7 all read 2 mV like PCB 1.

4) How does frequency affect the measured crosstalk for square vs. sine waves?
The noise from a square wave is not frequency dependent, but depends solely on the steep rise time, resulting in a large di/dt. This can be explained as the square wave consists of extremely high frequency sinusoids. The noise stemming from a sine-wave, however, is frequency dependent and the noise will increase with the frequency, while the square wave has the same amount of noise throughout a frequency range, this is as long as the rise time doesn't change.
This is only the case for badly designed boards, as a good design would keep crosstalk to a minimum even for square waves, where we would see no difference as we saw in the sweep test on a bad and good board, with the following results: The sine wave sweep gave us 2 mV → 6 mV → 73 mV from 1kHz to 10 MHz on PCB 4 while the square wave gave us 130 mV → 260 mV from 1kHz 10 MHz, this is because the second harmonic came into the measuring windows at higher frequencies. While a good board PCB 6 gave us 2 mV from 1kHz to 10 MHz both with a square and a sine wave, showing good design is important.

5) What does the board-to-board test demonstrate about electromagnetic coupling
A badly designed board can interfere with a well-designed one even without any electrical connection. Sliding/rotating the boards changes coupling because you're in the near field, where the mutual capacitance and inductance dominate. This shows even the layout and not just wires can affect the crosstalk and noise levels.

Plots of crosstalk vs frequency for a good and bad pcb



Note PCB4 square wave magnitude only doubles because both negative and positive peaks appear in the pk-pk window on the oscilloscope. Graph shows that a good board doesn't get affected by frequency, but a bad board gets worse, with higher frequencies, for sinus waves and allround bad for square waves.

Table showing board number, square crosstalk and sine crosstalk.

Board	Square Crosstalk [mV]	Sine Crosstalk [mV]	Notes
1	2 mV	2 mV	Noise floor ~2 mV
2	2 mV	2 mV	
3	29 mV	20 mV	Stronger leakage
4	40 mV	22 mV	
4+	261 mV	120 mV	Large coupling observed
5	2 mV	2 mV	No extra effect
6	2 mV	2 mV	
7	2 mV	2 mV	